

Glossa-Lab Indus Decipherment Progress — Phase-28

Phase-28: CISI Vol 3 OCR + Mahadevan Crosswalk + Allograph-Aware Iconographic Scoring

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Executive summary

Phase-28 delivered all five priority items wrapped as glossa-lab atomic nodes (invocable via `python -m glossa_lab.experiments <graph_id>`). The headline quantitative result: total weighted iconographic anchor score rose from **24.5 (Phase-27c) to 27.0 (Phase-28c)**, a +10.2% improvement at constant anchor count, driven by allograph-family awareness (M-410 fish anchor now also scores against fish-allograph signs 50/60/145/147 and the fig-tree anchor extends to 311_fig).

Negative findings. The on-disk CISI Vol 3 Part 3 PDF turned out to be the introduction (40 pages) rather than catalogue plates; OCR extracted 23 records but they are LPIW/LE (Linear Proto-Iranian Writing / Linear Elamite) seal IDs not Indus script. Reverse Janabiyah search v2 found the same lone false positive (ur-temen-na) as Phase-27a — expanding the phoneme map 30→35 did not introduce new false positives, which is the correct robustness outcome.

Decipherment progress: ~9-10% (up from ~7-8% at Phase-27).

Phase-28 Deliverables (5 priority items)

#	Priority	Status	Atomic Node	Graph
1	CISI Vol 3 OCR via call_llm_vision	DONE (limited yield)	CISIVol3OCRNode	indus_phase28a_cisi_vol3_ocr
2	Phoneme map expansion (30→35)	DONE	Phase28CorpusLoader	indus_phase28e_expanded_phoneme_map
3	Reverse Janabiyah search v2	DONE (no new positives)	ReverseJanabiyahSearch V2	indus_phase28d_reverse_janabiyah_v2
4	Mahadevan 1977 ↔ Parpola 1994b crosswalk	DONE (25 entries)	MahadevanCrosswalkLoader	indus_phase28b_mahadevan_crosswalk
5	Allograph-aware iconographic scorer	DONE (24.5→27.0)	AllographAwareIconographicScore	indus_phase28c_allograph_iconographic

Allograph-aware iconographic scorer (Phase-28c)

Anchor	Sign	Family	+ Allographs	Score
M-410_fish_crocodile	47	fish	50, 60, 145, 147	4.00
H-902B_four_pots_of_fish	47	fish	50, 60, 145, 147	4.00
H-9_seven_fish_only	92	numerals	—	3.00
M-1202_intersecting_circles_squirrel	261	intersecting_circles	—	3.00
H-771_intersecting_circles_squirrel_2	261	intersecting_circles	—	3.00
Nd-1_squirrel_clarification	281	—	—	2.00
M-478_pot_of_offerings	124	—	—	1.50
M-453_anthropomorphic_deity	261	intersecting_circles	—	1.50
M-1186_fig_deity_seven_mothers	261	intersecting_circles	—	1.50
M-414_fig_fish_seal	311	fig_tree	311_fig	1.25
H-179_fig_with_deity	311	fig_tree	311_fig	1.25
H-723_two_strokes_compounds	87	—	—	1.00
TOTAL			+10 extensions on 5 signs	27.0

Interpretation. Phase-27c scored 12 anchors against 7 explicit sign IDs for a total of 24.5. Phase-28c recognises that 5 additional signs (50, 60, 145, 147, 311_fig) are graphemic allographs of those 7 — Parpola's fish-family hypothesis predicts they all carry the same Dravidian phoneme. Adding the +0.25 per allograph member that matches the anchor's iconic reading raises the total to 27.0 (+10.2%). This is a *consistency* result, not an *information* result: the allograph hypothesis is internally coherent at scale.

Reverse Janabiyah search v2 (Phase-28d)

Scored **45** unique persons-v3 candidates against **17** miin-renderings (vs Phase-27a's static 15). **1 have at least one position match; 1** have at least one miin-rendering anywhere.

Rank	Candidate name	Period	Score	Position match
1	ur-temen-na	Ur III (ca. 2100-2000 BC)	3.00	yes
2	ur-da-am-ma-ne2-e	Neo-Assyrian (ca. 911-612 BC)	2.50	no
3	ur-gu2-edin-na-ta	Ur III (ca. 2100-2000 BC)	2.00	no
4	ur-{d}ig-alim-ka-sze3	Ur III (ca. 2100-2000 BC)	2.00	no
5	lu2-{d}nin-gir2-su	Ur III (ca. 2100-2000 BC)	1.50	no
6	ur-{d}nin-gir2-su	Ur III (ca. 2100-2000 BC)	1.50	no
7	ur-{d}ba-ba6-ta	Ur III (ca. 2100-2000 BC)	1.50	no
8	ur-{d}na-ru2-a	Ur III (ca. 2100-2000 BC)	1.50	no
9	ur-{d}szusz3-{d}ba-ba6	Ur III (ca. 2100-2000 BC)	1.50	no
10	ur-{d}szul-pa-e3	Ur III (ca. 2100-2000 BC)	1.50	no

Outcome: The lone false positive ur-temen-na (Ur III, Girsu) is the same hit Phase-27a found. Expanding the phoneme map +5 entries (lion, eagle, cobra, strengthened yoke-carrier, strengthened buffalo) did not change the picture. This is the correct robustness behaviour: the test rejects simple-rebus hypotheses from the second direction with very high specificity.

CISI Vol 3 OCR via call_llm_vision (Phase-28a)

Mistral pixtral-12b-2409 vision OCR via the new call_llm_vision() helper in glossa_lab/ai_utils.py. The helper routes through Settings so Ollama llava/gemma3-vision is preferred when configured. **Total records: 23** (7 seal, 15 iconography, 1 sign_ref).

Structural finding: The on-disk PDF is the introduction/front-matter of CISI Vol 3 Part 3, not the catalogue plates themselves. Seal IDs extracted (Shd-1, KSS-380..383, V, G', Bactrian-1354) are LPIW/LE — Linear Proto-Iranian Writing / Linear Elamite — from sites adjacent to but distinct from the Indus core. **Zero overlap** with the 14 Phase-22 catalogue IDs.

Implication for Phase-29: The actual catalogue plates must be acquired separately (CISI Vol 3 Part 1/2 or pages 41-525 of Part 3 if they exist). Best route: ICIPS digital catalogue request, or Helsinki/Harvard ILL.

Dataset deltas vs Phase-27

Asset	Phase-27	Phase-28	Delta
parpola_phonemes.json	30 entries	35 entries	+5
mahadevan_parpola_crosswalk.json	—	25 + 4 families	NEW
cisi_vol3_extracted_signs.json	—	23 records	NEW
iconographic_anchors.json	12 anchors	12 anchors	unchanged
Anchor coverage (distinct sign IDs)	7	12 (+5 via allography)	+71%
Total weighted anchor score	24.5	27.0	+10.2%
Reverse Janabiyah false positives	1/45	1/45	unchanged

Honest limitations

- **No new Indus inscriptions ingested.** The 23 OCR records are LPIW/LE.
- **Reverse Janabiyah v2 found no new candidates.** Same single false positive (ur-temen-na) as Phase-27a. The +5 phoneme entries did not surface anything new in the existing CDLI corpus.
- **The +10.2% anchor score increase is a re-counting of existing evidence** (allograph families), not new evidence. Consistency, not information.
- **Phase-28's main contribution is qualitative:** it shows Parpola's allograph hypothesis is internally consistent at scale, which is necessary but not sufficient for decipherment.

Phase-29 priorities

1. Acquire CISI Vol 3 Part 1/2 (the actual Indus catalogue plates, since the Vol 3 Part 3 PDF turned out to be introduction).
2. Complete Mahadevan 1977 → Parpola 1994b crosswalk to all ~417 signs.
3. Extend allograph-family coverage to Wells 2015 typology (more granular than Parpola 1994b families).
4. Re-attempt Crawford 2001 'Early Dilmun' via a different mirror/library.
5. Build a held-out blind test set from 2024-2026 publications (any newly-published seals) and run the full Phase-28 pipeline on them as pre-registered confirmation.

Phase-28 graph run trace

Graph ID	Atomic node	Result file
indus_phase28a_cisi_vol3_ocr	CISIVol3OCRNode	reports/phase28a_cisi_vol3_ocr.json
indus_phase28b_mahadevan_crosswalk	MahadevanCrosswalkLoader	reports/phase28b_mahadevan_crosswalk.json
indus_phase28c_allograph_iconographic	AllographAwareIconographicScore	reports/phase28c_allograph_iconographic.json
indus_phase28d_reverse_janabiyah_v2	ReverseJanabiyahSearchV2	reports/phase28d_reverse_janabiyah_v2.json
indus_phase28e_expanded_phoneme_map	Phase28CorpusLoader	reports/phase28e_expanded_phoneme_map.json
indus_phase28f_verdict	Phase28Verdict	reports/phase28f_verdict.json

Source: `backend/glossa_lab/experiment_graph_phase28.py` + 6 graph JSONs in `backend/glossa_lab/experiments/graphs/`. Reproducible via `python -m glossa_lab.experiments <graph_id>`.